Dario Serrano

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Portfolio: lizmutt.github.io/darioserrano.github.io/
U.S. Citizen (U.S. Person) • EU Citizen (Finland) • Canadian Work Permit — **no visa sponsorship required**
Native fluency in **English** and **Mandarin**; also Shanghainese

Education

University of Toronto 2023–2027
B.A.Sc. Mechanical Engineering — Minor: Robotics & Mechatronics
Certificates: Engineering Business, Artificial Intelligence Engineering
Coursework: Robotics (AER525), Control Systems (MIE404), Machine Learning (ECE421), Embedded Systems (MIE438)

Research & Engineering Experience

Flight Control Engineering Intern — HopTo Robotics Jun–Aug 2026
Shenzhen, China / Hong Kong

- Developed flight control and hardware for a novel water-hopping quadrotor that recovers energy from the water surface via a hydrofoil; work forms the basis of an IROS submission.
- Contributed mechanical design and bench testing to the hopping-robot platform commercialized from Prof. Chirarattananon's research; represented the company at the IEEE ICRA 2026 exhibition in Vienna, presenting hardware in English and Mandarin.

Undergraduate Researcher — CLIMB Lab, University of Toronto Sep 2026–Present
Advisor: Prof. Pakpong Chirarattananon

- Leading flight control research on a revolving-wing quadrotor as first author with a senior PhD collaborator; running experiments in the lab's motion-capture facility to characterize performance against ground truth.

Team Lead — MIE491 Capstone, Ice-Perching Micro Air Vehicle Sep 2026–Apr 2027
Client: Synetic Labs — Supervisor: Prof. Pakpong Chirarattananon

- Coordinating a three-person team designing a MAV-scale mechanism that perches and latches onto steep ice, extending published work from larger platforms down to micro scale.
- Leading flight control development and R&D of the novel perching mechanism: scaling analysis, compliant-mechanism design, and a drop-test rig for impact and holding-force characterization.

Suspension Team — UofT Baja SAE Racing Sep 2025–Present
Toronto, ON

- Designed front suspension A-arms and the brake hub to steering knuckle interface in SolidWorks for the team's first vehicle in six years, built and completed in 2026; performed MIG/TIG welding and manual machining of suspension and chassis components to competition tolerances. Team is registered for 2027 competition.

Engineering Intern — Schindler Group Jul–Aug 2024
Shanghai, China

- Modeled elevator structural components in Creo Parametric with the worldwide R&D team and performed FEA to FKM guidelines; built MATLAB ODE models to validate simulation accuracy.

Publications

1. **A Thrust-Assisted Cable-Driven Jumping and Hopping Robot**. S. Bai, . . . , **D. Serrano**, P. Chirarattananon. *Under review, IEEE Transactions on Robotics.*
2. **Water-Hopping Quadrotor** (working title). X. Lian*, **D. Serrano***, F. Wang*, P. Chirarattananon. *In process of submission to IEEE/RSJ IROS 2027.* *Equal contribution.
3. **Revolving-Wing Quadrotor Flight Control** (working title). **D. Serrano**, S. Li, P. Chirarattananon. *In preparation, targeting IEEE Robotics and Automation Letters.*

Selected Projects

Seizure Detection with CNN–LSTM — Deep Learning May–Sep 2025

- Built a CNN–LSTM in PyTorch for EEG seizure classification (CHB-MIT) with Butterworth filtering, wavelet denoising and sliding-window segmentation; AUROC 0.96, F1 0.84 on unseen patients.

Current-Sensing & Anti-Aliasing PCB — Analog and Digital Circuits Jan–May 2026

- Designed an analog current-sensing front end (LM324) and a 3rd-order Chebyshev LPF (1 kHz cutoff, $\geq$35 dB at 5 kHz) for 50–70 kHz ADC sampling, validated in LTspice; laid out a 2-layer board in EAGLE and hand-soldered the prototype.

Technical Skills

Design & Fabrication: SolidWorks, Creo, ANSYS (FEA), EAGLE (PCB), LTspice, 3D printing, machining, MIG/TIG welding
Programming: Python, PyTorch, MATLAB/Simulink, NumPy, pandas, C (embedded)